CSA5108 Cryptography and Network security

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Exp 6: Breaking an Affine Cipher
Code:

```
#include <stdio.h>
#include <ctype.h>

int modInverse(int a) {
   for (int i = 0; i < 26; i++)
      if ((a * i) % 26 == 1)
         return i;
   return -1;
}

int main() {
   char cipher[5000];
   int a = 3, b = 15;
   int inv;

   printf("Enter ciphertext: ");
   fgets(cipher, sizeof(cipher), stdin);

   inv = modInverse(a);

   for (int i = 0; cipher[i] != '\0'; i++) {
      if (isupper(cipher[i])) {
         int c = cipher[i] - 'A';
         int p = (inv * (c - b + 26)) % 26;
         cipher[i] = p + 'A';
      }
      else if (islower(cipher[i])) {
         int c = cipher[i] - 'a';
         int p = (inv * (c - b + 26)) % 26;
         cipher[i] = p + 'a';
```

```
        }
    }

    printf("Decrypted text: %s", cipher);

    return 0;
}
```

